

Crop Cultivation Guide

AgriMitra AI Knowledge Base — Document 1 of 6 — Major Crop Practices

Paddy (Rice)

Overview

Staple cereal crop, grown widely across irrigated deltas and canal-fed regions such as the Godavari and Krishna delta areas of Andhra Pradesh.

Suitable Soil & Climate

- Best suited to clayey loam or alluvial soils with good water retention
- Requires standing water for most of the growth period
- Optimum temperature range: 20–35°C; needs high humidity

Sowing Season & Method

- Two main seasons: Kharif (June–July sowing) and Rabi (November–December sowing)
- Nursery raising followed by transplanting at 20–25 days old seedlings is the common method
- Direct seeding is also practiced in water-scarce areas
- Recommended spacing: 20 cm x 15 cm between transplanted seedlings

Fertilizer Schedule

Split application of Nitrogen at basal, tillering, and panicle initiation stages. See Document 2 (Fertilizer & Nutrient Management) for stage-wise schedule.

Common Pests & Diseases

Watch for stem borer, brown planthopper, and blast disease. See Documents 3 and 4 for detailed pest and disease management.

Irrigation Requirement

- Maintain 2–5 cm standing water during vegetative and reproductive stages
- Drain field 7–10 days before harvest to aid ripening and easy harvesting

Harvesting Guidelines

- Harvest when 80–85% of grains turn golden yellow
- Grain moisture at harvest should be around 20–22% for good milling quality
- Dry harvested grain to 12–14% moisture before storage

Note for verification: Sowing windows and durations vary by local agro-climatic zone. Cross-check with your district Krishi Vigyan Kendra (KVK) or state agriculture department before advising exact dates.

Cotton

Overview

Major commercial fibre crop grown in black cotton soils (regur) across the Rayalaseema and coastal districts.

Suitable Soil & Climate

- Deep black cotton soil (regur) with good moisture-holding capacity is ideal
- Requires a frost-free period of at least 180 days
- Optimum temperature: 21–30°C during growth period

Sowing Season & Method

- Sown at the onset of monsoon, typically May–June (rainfed) or April–May (irrigated)
- Recommended spacing: 90 cm x 60 cm for hybrid varieties (adjust per local recommendation)
- Seed treatment with recommended fungicide/insecticide before sowing is advised

Fertilizer Schedule

Requires balanced N-P-K with special attention to Potassium for fibre quality. See Document 2 for the stage-wise schedule.

Common Pests & Diseases

Bollworms (pink, American, spotted) and sucking pests (aphids, jassids, whitefly) are major concerns. See Documents 3 and 4.

Irrigation Requirement

- Critical irrigation stages: squaring, flowering, and boll development
- Avoid water stress during flowering and boll formation to prevent shedding

Harvesting Guidelines

- Picking begins when bolls fully open, typically in 2–3 rounds, 15–20 days apart
- Pick in dry weather to maintain fibre quality and avoid staining

Note for verification: Sowing windows and durations vary by local agro-climatic zone. Cross-check with your district Krishi Vigyan Kendra (KVK) or state agriculture department before advising exact dates.

Chilli

Overview

High-value commercial spice crop, with Andhra Pradesh being one of India's leading chilli-producing states.

Suitable Soil & Climate

- Well-drained sandy loam to red loamy soils are preferred
- Sensitive to waterlogging; requires good drainage
- Optimum temperature: 20–30°C

Sowing Season & Method

- Nursery sown in May–June, transplanted 30–35 days later for Kharif crop
- Rabi crop nursery sown in October–November
- Recommended spacing: 60 cm x 45 cm

Fertilizer Schedule

Responds well to organic manure combined with balanced N-P-K; avoid excess Nitrogen, which encourages foliage over fruiting. See Document 2.

Common Pests & Diseases

Thrips, mites, and fruit rot/anthracnose (die-back) are the major yield-limiting factors. See Documents 3 and 4.

Irrigation Requirement

- Light, frequent irrigation is preferred over heavy flooding
- Critical stages: flowering and fruit development
- Drip irrigation is well-suited and widely recommended for chilli

Harvesting Guidelines

- Green chilli picking starts 60–75 days after transplanting
- For dry red chilli, harvest when pods turn fully red and begin to wrinkle

Note for verification: Sowing windows and durations vary by local agro-climatic zone. Cross-check with your district Krishi Vigyan Kendra (KVK) or state agriculture department before advising exact dates.

Maize

Overview

Versatile cereal crop grown in both Kharif and Rabi seasons, used for food, feed, and industrial purposes.

Suitable Soil & Climate

- Well-drained sandy loam to clay loam soils with good organic matter
- Sensitive to waterlogging, especially at early growth stages
- Optimum temperature: 21–27°C

Sowing Season & Method

- Kharif sowing: June–July; Rabi sowing: October–November
- Recommended spacing: 60 cm x 20 cm
- Seed rate approximately 20–25 kg/hectare depending on variety

Fertilizer Schedule

Heavy Nitrogen feeder; requires split doses at basal, knee-high, and tasseling stages. See Document 2.

Common Pests & Diseases

Fall armyworm, stem borer, and turicum leaf blight are key concerns. See Documents 3 and 4.

Irrigation Requirement

- Critical stages: knee-high, tasseling/silking, and grain filling
- Avoid moisture stress during tasseling — most sensitive stage for yield loss

Harvesting Guidelines

- Harvest when husk turns straw-colored and kernels harden (black layer formation)
- Grain moisture at harvest typically 20–25%; dry to 12–14% for safe storage

Note for verification: Sowing windows and durations vary by local agro-climatic zone. Cross-check with your district Krishi Vigyan Kendra (KVK) or state agriculture department before advising exact dates.

Groundnut

Overview

Major oilseed and legume crop, well suited to the sandy and red loamy soils of Rayalaseema region.

Suitable Soil & Climate

- Well-drained sandy loam soils with good calcium content are ideal (important for pod development)
- Optimum temperature: 25–30°C

Sowing Season & Method

- Kharif sowing: June–July; Rabi/summer sowing: January–February (irrigated)
- Recommended spacing: 30 cm x 10 cm for bunch varieties
- Seed treatment with Rhizobium and PSB culture improves nodulation

Fertilizer Schedule

Being a legume, requires modest Nitrogen but responds strongly to Phosphorus, Calcium (gypsum), and Sulphur. See Document 2.

Common Pests & Diseases

Leaf miner, Spodoptera, and tikka leaf spot / rust are common. See Documents 3 and 4.

Irrigation Requirement

- Critical stages: flowering, pegging, and pod development
- Avoid water stress during pegging — directly affects pod set

Harvesting Guidelines

- Harvest when leaves start yellowing and pods show mature veining/dark inner shell color
- Dry pods to 8–10% moisture before storage

Note for verification: Sowing windows and durations vary by local agro-climatic zone. Cross-check with your district Krishi Vigyan Kendra (KVK) or state agriculture department before advising exact dates.

Sugarcane

Overview

Long-duration cash crop grown in canal-irrigated tracts, particularly in coastal Andhra Pradesh.

Suitable Soil & Climate

- Deep, well-drained loamy soils with good water-holding capacity
- Requires a long frost-free growing period (10–12 months)
- Optimum temperature: 20–35°C

Sowing Season & Method

- Main planting season: December–February (Adsali) or June–July (Kharif)
- Planted using healthy 2–3 budded setts
- Recommended row spacing: 90–120 cm

Fertilizer Schedule

Very high nutrient demand; needs heavy Nitrogen and Potassium in split doses across the long growth cycle. See Document 2.

Common Pests & Diseases

Early shoot borer, top borer, and red rot disease are major threats. See Documents 3 and 4.

Irrigation Requirement

- Requires frequent irrigation, especially during the grand growth phase
- Drip irrigation is increasingly recommended to improve water-use efficiency

Harvesting Guidelines

- Harvest at 10–12 months when cane shows maximum sugar (brix) content
- Harvest close to the ground to maximize yield from the base of the cane

Note for verification: Sowing windows and durations vary by local agro-climatic zone. Cross-check with your district Krishi Vigyan Kendra (KVK) or state agriculture department before advising exact dates.